



GIRRAWEE HIGH SCHOOL

MATHEMATICS EXTENSION

YEAR 11

TASK 3

- **Instructions**

- Time Allowed – 1 hour.
- Write using black pen.
- Calculators approved by NESA may be used
- This task is open book but NOT open technology (ICT).
- Show all relevant mathematical reasoning and/or calculations.
- Marks may be deducted for careless or badly arranged work.

You must have your

- camera on (no virtual backgrounds)
- microphone off
- speakers on
- phone on the desk facing upwards and on silent and be visible to the teachers

Show all relevant mathematical reasoning and/or calculations.

Question 1 (15 marks)

a. Consider the polynomial $P(x) = 2x^3 + 3x^2 - 29x - 60$.

- (i) Find the remainder when $P(x)$ is divided by $(x + 2)$. [1]
- (ii) Use the factor theorem to show that $(x + 3)$ is a factor of $P(x)$. [1]
- (iii) Factorise $P(x)$ completely. [2]

b. If α , β and γ are the roots of $2x^3 + 3x^2 - 5x + 1 = 0$ find:

- (i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ [2]
- (ii) $\alpha^2 + \beta^2 + \gamma^2$ [2]
- (iii) $(\alpha - \beta)^2\gamma + \alpha(\beta - \gamma)^2 + (\alpha - \gamma)^2\beta$ [2]

c. The polynomial $P(x) = ax^4 + bx^3 + cx^2 + e$

has a remainder -3 when divided by $x - 1$.

The polynomial has a double root at $x = -1$.

Find the value of $4a + 2c$. [2]

d. Let $P(x) = (x + 1)(x - 3)Q(x) + a(x + 1) + b$, where $Q(x)$ is a polynomial and a and b are real numbers.

When $P(x)$ is divided by $(x + 1)$ the remainder is -11 .

When $P(x)$ is divided by $(x - 3)$ the remainder is 1 .

- (i) Find the value of b [1]
- (ii) What is the remainder when $P(x)$ is divided by $(x + 1)(x - 3)$? [2]

Question 2 (15 marks)

- a. In how many ways can the letters of the word **DEDUCED** be arranged? [2]
- b. How many numbers greater than 3 000 can be formed from the digits 1, 3, 5, 7, 9 if no repetition is permitted? [2]
- c. In how many ways can a committee of 3 women and 4 men be chosen from 8 women and 7 men if two particular women cannot serve on the committee together? [3]
- d. Mr and Mrs Antony (the hosts) and 6 guests sit around a circular dining table. In how many ways can they be arranged if the two hosts are separated? [2]
- e. At a fund-raising dinner, each rectangular table has 9 seats, 5 facing the stage and 4 with their backs to the stage. In how many ways can nine people be seated at a particular table if:
 - (i) Angelo and Agam must sit on the same side? [2]
 - (ii) Angelo and Agam must sit on opposite sides? [2]
- f. A box contains 2 red, 3 blue, 9 green and 9 yellow balls. What is the minimum number of balls that must be drawn to be certain that there will be 4 balls of the same colour? [2]

Question 3 (15 marks)

a. Simplify $\frac{5^n + 5^{n+2}}{5^{n+1}}$ [2]

b. Solve for x : $2\log_7 5 = \log_7 2x - \log_7 3$ [2]

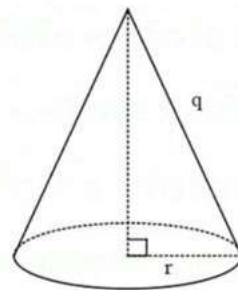
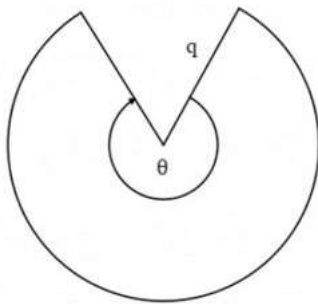
c. A function is described by the parametric equations

$$x = t + 1; y = t^2 - 1$$

Find the cartesian equation of this function. [2]

d. A sector with angle θ with fixed radius q is folded to form a cone with radius r , as shown below.

Let the area of the sector be A and the area of the base of the cone be B .



(i) Show that $r = \frac{q\theta}{2\pi}$. [1]

(ii) Find the exact value of θ such that $3A^2 = 4B^2$. [2]

e. Solve:

(i) $2 \cos (2\theta - 30^\circ) = \sqrt{3}$; $0^\circ \leq \theta \leq 360^\circ$ [3]

(ii) $2 \log_2 (x + 5) - \log_2 (x + 9) = 1$ [3]

End of Examination

2021 GHS Year 11 Mathematics Ext Task 3
SOLUTIONS

Question 1

a) $P(x) = 2x^3 + 3x^2 - 29x - 60$

i) $P(-2) = 2(-2)^3 + 3(-2)^2 - 29(-2) - 60$
 $= -6$

$\therefore$ Remainder is -6 .

ii) $P(-3) = 2(-3)^3 + 3(-3)^2 - 29(-3) - 60$
 $= 0$

$\therefore (x+3)$ is a factor of $P(x)$.

iii)

$$\begin{array}{r}
 2x^2 - 3x - 20 \\
 x+3 \overline{) 2x^3 + 3x^2 - 29x - 60} \\
 \underline{- 2x^3 + 6x^2} \\
 -3x^2 - 29x \\
 \underline{- -3x^2 - 9x} \\
 -20x - 60 \\
 \underline{- -20x - 60} \\
 0
 \end{array}$$

$\therefore 2x^3 + 3x^2 - 29x - 60 = (x+3)(2x^2 - 3x - 20)$

$\therefore P(x) = (x+3)(2x+5)(x-4)$

b) $2x^3 + 3x^2 - 5x + 1 = 0$

i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$

$= \frac{-5/2}{-1/2} = 5$

$$\begin{aligned}
 \text{ii) } \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) \\
 &= \left(-\frac{3}{2}\right)^2 - 2\left(-\frac{5}{2}\right) \\
 &= \frac{29}{4} = 7\frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) } (\alpha\beta)^2\gamma + \alpha(\beta\gamma)^2 + (\alpha\gamma)^2\beta \\
 &= \alpha\beta\gamma(\alpha\beta + \beta\gamma + \alpha\gamma) \\
 &= \left(-\frac{1}{2}\right)\left(-\frac{5}{2}\right) \\
 &= \frac{5}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } P(x) &= ax^4 + bx^3 + cx^2 + e \\
 P'(x) &= 4ax^3 + 3bx^2 + 2cx
 \end{aligned}$$

$$P(1) = a + b + c + e = -3 \quad \text{--- ①}$$

$$P(-1) = a - b + c + e = 0 \quad \text{--- ②}$$

$$P'(-1) = -4a + 3b - 2c = 0$$

$$\therefore 4a + 2c = 3b \quad \text{--- ③}$$

$$\text{①} - \text{②}$$

$$2b = -3$$

$$b = -\frac{3}{2}$$

substitute $b = -\frac{3}{2}$ into ③

$$4a + 2c = 3\left(-\frac{3}{2}\right)$$

$$\therefore 4a + 2c = -\frac{9}{2}$$

$$d) P(x) = (x+1)(x-3)Q(x) + a(x+1) + b$$

$$i) P(-1) = (0)(-4)Q(x) + a(0) + b = -11$$

$$\therefore b = -11$$

$$ii) P(3) = (0)Q(x) + 4a + b = 1$$

$$4a + b = 1$$

$$4a - 11 = 1$$

$$4a = 12$$

$$a = 3$$

When $P(x)$ is divided by $(x+1)(x-3)$

$$R(x) = a(x+1) + b$$

$$= 3(x+1) - 11$$

$$= 3x + 3 - 11$$

$$= 3x - 8$$

Question 2

a) DEDUCED

7 letters

3 Ds

2 Es

$$= \frac{7!}{3!2!}$$

$$= 420 \text{ ways.}$$

b) 1, 3, 5, 7, 9 ; Numbers > 3000

$$5 \text{ digit numbers} = 5! = 120$$

$$4 \text{ digit numbers} = 4 \times 4!$$

$$= 96$$

$$\text{Total} = 120 + 96$$

$$= 216 \text{ numbers.}$$

c) only 1 of the 2 particular woman chosen

$${}^6C_2 \times 2 \times {}^7C_4$$

$$= 1050$$

both of the women not chosen.

$${}^6C_3 \times {}^7C_4$$

$$= 700$$

$$\text{Total ways} = 1750$$

d) Hosts separated = Total — hosts together.

$$= 7! - (1 \times 2 \times 6!)$$

$$= 3600$$

$$(\text{OR}) \quad 1 \times 5 \times 6! = 3600$$

e) i) same side

$$\text{Both facing stage} : {}^5P_2 \times 7!$$

$$\text{Both not facing} : {}^4P_3 \times 7!$$

$$\text{Both on the same side} = 161280$$

ii) on opposite sides :

$$= 2 \left[{}^5P_1 \times {}^4P_1 \times 7! \right]$$

$$= 201600$$

$$f) \quad 2R, 3B, 9G, 9Y$$

Minimum number drawn to guarantee
4 balls of same colour.

$$\begin{aligned} 2 + 3 + 3 + 3 + 1 \\ = 12 \end{aligned}$$

Question 3

$$a) \quad \frac{5^n + 5^{n+2}}{5^{n+1}}$$

$$= \frac{5^n (1 + 5^2)}{5^n \times 5^1}$$

$$= \frac{26}{5}$$

$$b) \quad 2 \log_7 5 = \log_7 2x - \log_7 3$$

$$\log_7 (5^2) = \log_7 \left(\frac{2x}{3} \right)$$

$$25 = \frac{2x}{3}$$

$$x = \frac{75}{2}$$

$$c) \quad x = t + 1 \quad \text{--- (1)} ; \quad y = t^2 - 1 \quad \text{--- (2)}$$

$$\Rightarrow t = x - 1$$

Substitute $t = x - 1$ into (2)

$$y = (x - 1)^2 - 1$$

$$= x^2 - 2x + 1 - 1$$

$$y = x^2 - 2x$$

d) i)

the arc length of the sector will become the circumference of the base of the cone.

$$\therefore r\theta = 2\pi r$$

$$r = \frac{r\theta}{2\pi}$$

ii)

$$3A^2 = 4B^2$$

$$A = \frac{1}{2} r^2 \theta ; \quad B = \pi r^2$$

$$r = \frac{r\theta}{2\pi} \Rightarrow B = \pi \left(\frac{r\theta}{2\pi} \right)^2$$

$$= \frac{r^2 \theta^2}{4\pi}$$

$$3A^2 = 3 \left(\frac{1}{2} r^2 \theta \right)^2$$

$$= \frac{3r^4 \theta^2}{4}$$

$$4B^2 = 4 \left(\frac{q^2 \theta^2}{4\pi} \right)^2$$

$$= \frac{q^4 \theta^4}{4\pi^2}$$

Solving $3A^2 = 4B^2$

$$\frac{3q^4 \theta^2}{4} = \frac{q^4 \theta^4}{4\pi^2}$$

$$3\pi^2 = \theta^2$$

$$\theta = \sqrt{3} \pi ; \theta > 0$$

e) $2 \cos(2\theta - 30^\circ) = \sqrt{3} ; 0^\circ \leq \theta \leq 360^\circ$

$$\cos(2\theta - 30^\circ) = \frac{\sqrt{3}}{2}$$

$$2\theta - 30^\circ = -30^\circ, 30^\circ, 330^\circ, 390^\circ, 690^\circ$$

$$2\theta = 0^\circ, 60^\circ, 360^\circ, 420^\circ, 720^\circ$$

$$\theta = 0^\circ, 30^\circ, 180^\circ, 210^\circ, 360^\circ$$

$$0^\circ \leq 2\theta \leq 720^\circ$$

$$-30^\circ \leq 2\theta - 30^\circ \leq 690^\circ$$

$$4) \quad 2\log_2(x+5) - \log_2(x+9) = 1$$

$$\log_2(x+5)^2 - \log_2(x+9) = 1$$

$$\log_2\left(\frac{(x+5)^2}{x+9}\right) = 1$$

$$\frac{(x+5)^2}{x+9} = 2$$

$$(x+5)^2 = 2(x+9)$$

$$x^2 + 10x + 25 = 2x + 18$$

$$x^2 + 8x + 7 = 0$$

$$(x+7)(x+1) = 0$$

$$x = -7, -1.$$

$$\therefore x = -1 \quad ; \quad \text{since } x > -5$$